

Actividad 02

Nmap

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Realizare el ejercicio de Nmap con mi propia maquina virtual de kali al momento de scanear mi IP es **10.0.2.15**

```
Session Actions Edit View Help
(kali㉿kali)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 82789sec preferred_lft 82789sec
    inet6 fd17:625c:f037:2:ca50:3329:bc45:ca6/64 scope global dynamic noprefixroute
        valid_lft 86392sec preferred_lft 14392sec
    inet6 fe80::9718:9793:3bfa:cff0/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Usaremos el nmap scanme.nmap.org para ver si esta instalado

```
(kali㉿kali)-[~]
$ nmap scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 14:50 -0400
Stats: 0:00:02 elapsed; 0 hosts completed (0 up), 0 undergoing Host Discovery
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.016s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http

Nmap done: 1 IP address (1 host up) scanned in 10.35 seconds
```

Escaneamos nuestro objetivo 192.168.100.14 (mi propia computadora) y parece ser que solo tenemos los servicios de 135 mapea el sistema operativo, 139 netbios y 445 en este puerto se comparten archivos con protocolo TCP/IP.

```
(kali@kali) [~]
$ nmap 192.168.100.14
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 21:35 -0400
Nmap scan report for 192.168.100.14
Host is up (0.0016s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
Nmap done: 1 IP address (1 host up) scanned in 5.10 seconds
```

Buscaremos si Nuestro Host esta active con nmap -sn 192.168.100.14 y nos dice que si esta activa

```
(kali@kali) [~]
$ nmap -sn 192.168.100.14
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 21:53 -0400
Nmap scan report for 192.168.100.14
Host is up (0.00087s latency)
Nmap done: 1 IP address (1 host up) scanned in 2.04 seconds
```

Con nmap -p- buscaremos si hay mas puertos TCP abiertos (duro muchisimo)

```
(kali@kali)-[~]
└─$ nmap -p- 192.168.100.14
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 21:57 -0400
Stats: 0:09:17 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth S
SYN Stealth Scan Timing: About 54.06% done; ETC: 22:14 (0:07:53 remaining)
Stats: 0:10:34 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth S
SYN Stealth Scan Timing: About 54.19% done; ETC: 22:17 (0:08:56 remaining)
Stats: 0:13:41 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth S
SYN Stealth Scan Timing: About 27.75% done; ETC: 22:46 (0:35:38 remaining)
Stats: 0:15:03 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth S
SYN Stealth Scan Timing: About 21.13% done; ETC: 23:08 (0:56:10 remaining)
Stats: 0:18:11 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth S
SYN Stealth Scan Timing: About 21.18% done; ETC: 23:23 (1:07:40 remaining)
Stats: 0:21:15 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth S
SYN Stealth Scan Timing: About 15.78% done; ETC: 00:12 (1:53:20 remaining)
Warning: 192.168.100.14 giving up on port because retransmission cap hit (10
Stats: 0:26:26 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth S
SYN Stealth Scan Timing: About 15.85% done; ETC: 00:44 (2:20:19 remaining)
Stats: 0:28:57 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth S
SYN Stealth Scan Timing: About 15.89% done; ETC: 00:59 (2:33:14 remaining)
Stats: 0:32:03 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth S
SYN Stealth Scan Timing: About 15.94% done; ETC: 01:18 (2:48:58 remaining)
Stats: 0:39:42 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth S
SYN Stealth Scan Timing: About 16.04% done; ETC: 02:05 (3:27:51 remaining)
Stats: 0:39:56 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth S
SYN Stealth Scan Timing: About 16.04% done; ETC: 02:06 (3:29:02 remaining)
Stats: 0:40:09 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth S
can
SYN Stealth Scan Timing: About 16.04% done; ETC: 02:07 (3:30:07 remaining)
```

Para verificar solo algunos puertos fui mas especifica, envíe un commando verificando solo del servicio 22 ssh, 80 http (que sabemos que esta abierto) y el 443 que es el Https donde el 22 y el 443 aparecen que estan filtered, esto no quiere decir que el Puerto este Cerrado, quiere decir que no hay suficiente informacion porque algo lo esta bloqueando, como lo muestra en esta imagen donde hice un test con mi IP de Kali

```

(kali@kali)-[~]
$ nmap -p 22,80,443 10.0.2.15
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 16:07 -0400
Nmap scan report for 10.0.2.15
Host is up (0.000073s latency).

PORT      STATE      SERVICE
22/tcp    filtered  ssh
80/tcp    open      http
443/tcp   filtered  https

Nmap done: 1 IP address (1 host up) scanned in 3.77 seconds

(kali@kali)-[~]
$ nmap --top-ports 100 10.0.2.15
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 16:09 -0400
Nmap scan report for 10.0.2.15
Host is up (0.000023s latency)
Not shown: 99 filtered tcp ports (no-response)
PORT      STATE      SERVICE
80/tcp    open      http

Nmap done: 1 IP address (1 host up) scanned in 6.28 seconds

```

Y ahora lo hice con la IP de mi computadora, parece ser que salen muchísimos mas resultados que en su mayoría salen filtered

```

(kali@kali)-[~]
$ nmap -p 22,80,443 192.162.100.14
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 22:53 -0400
Nmap scan report for vps1766.maxhosting.ru (192.162.100.14)
Host is up (0.00051s latency).

PORT      STATE      SERVICE
22/tcp    filtered  ssh
80/tcp    filtered  http
443/tcp   filtered  https

Nmap done: 1 IP address (1 host up) scanned in 3.24 seconds

```

También se usó nmap --top-ports y en su mayoría también están filtered

```
(kali@kali)-[~]
└─$ nmap --top-ports 22,80,443 192.162.100.14
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 22:56 -0400
Nmap scan report for vps1766.maxhosting.ru (192.162.100.14)
Host is up (0.00076s latency).

PORT      STATE SERVICE
21/tcp    filtered ftp
22/tcp    filtered ssh
23/tcp    filtered telnet
25/tcp    filtered smtp
53/tcp    filtered domain
80/tcp    filtered http
110/tcp   filtered pop3
111/tcp   filtered rpcbind
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
143/tcp   filtered imap
443/tcp   filtered https
445/tcp   filtered microsoft-ds
587/tcp   filtered submission
993/tcp   filtered imaps
995/tcp   filtered pop3s
1025/tcp  filtered NFS-or-IIS
1723/tcp  filtered pptp
3306/tcp  filtered mysql
3389/tcp  filtered ms-wbt-server
5900/tcp  filtered vnc
8080/tcp  filtered http-proxy
```

Ahora tratemos de verificar con nmap – F 192.162.100.14, donde aparecen abiertos los puertos 199 y el 873, en ellos podríamos adentrarnos

```
(kali@kali)-[~]
└─$ nmap -F 192.162.100.14
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 23:16 -0400
Nmap scan report for vps1766.maxhosting.ru (192.162.100.14)
Host is up (0.070s latency).
Not shown: 98 filtered tcp ports (no-response)
PORT      STATE SERVICE
199/tcp   open  smux
873/tcp   open  rsync

Nmap done: 1 IP address (1 host up) scanned in 7.28 seconds
```

Verificamos que servicios estan corriendo con nmap -Sv 192.162.100.14.

```

(kali@kali)-[~]
$ nmap -sV 192.162.100.14
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 23:34 -0400
Nmap scan report for vps1766.maxhosting.ru (192.162.100.14)
Host is up (0.026s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
199/tcp   open  smux    Linux SNMP multiplexer
873/tcp   open  rsync   (protocol version 31)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap
.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 22.40 seconds

```

```

(kali@kali)-[~]
$ nmap -p 199 -sV -sC 192.162.100.14
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-20 00:08 -0400
Nmap scan report for vps1766.maxhosting.ru (192.162.100.14)
Host is up (0.043s latency).
PORT      STATE SERVICE VERSION
199/tcp   open  smux    Linux SNMP multiplexer
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap
.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.17 seconds

```

En el 199 podemos ver que el nmap sabe que estamos interactuando con un ambiente virtual por el vps. Ahora vamos a ver que hay en el otro 873 es un protocolo que funciona para hacer transferencias de archivos Podemos observar que viene con una version 31, esto quiere decir que es la ultima version.

```

(kali@kali)-[~]
$ nmap -p 873 -sV -sC 192.162.100.14
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-20 00:19 -0400
Nmap scan report for vps1766.maxhosting.ru (192.162.100.14)
Host is up (0.045s latency).
PORT      STATE SERVICE VERSION
873/tcp   open  rsync   (protocol version 31)

Service detection performed. Please report any incorrect results at https://
.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.14 seconds

```

Hice Tambien un escaneo de las UDPs que estuvieran abiertas y el unico que salio fue el 161, que coincide con el mismo servicio que corte en el Puerto TCP 873

```
(kali@kali)~$ sudo nmap -sU 192.162.100.14
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-20 04:32 -0400
Nmap scan report for vps1766.maxhosting.ru (192.162.100.14)
Host is up (0.017s latency).
Not shown: 999 open|filtered udp ports (no-response)
PORT      STATE SERVICE
161/udp   open  snmp
Nmap done: 1 IP address (1 host up) scanned in 15.87 seconds
```

Con un comando de scrip analizamos vulnerabilidades, y nos menciona esto dos ultimos port TCP abiertos 199 y el 873

```
(kali@kali)~$ nmap --script vuln 192.162.100.14
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-20 21:40 -0400
Nmap scan report for vps1766.maxhosting.ru (192.162.100.14)
Host is up (0.035s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
199/tcp   open  smux
873/tcp   open  rsync
Nmap done: 1 IP address (1 host up) scanned in 21.35 seconds
```

Detectar el Sistema operativo, donde Podemos observar que alcanza a visualizar la maquina virtual ademas de algunos servicios de Microsoft, creo que en si no podriamos decir con esto que es podria ser Windows.

```
(kali@kali)~$ nmap -O 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-21 01:43 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0010s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox-cpe:/o:ardamir:gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.45 seconds
```

Y para tener un poco mas de objetividad en el Sistema operativo utilizamos este comandito y nos confirma que si es un windows

```
(kali@kali) ~$ sudo nmap -A 192.168.56.1
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-21 01:48 -0400
Nmap scan report for 192.168.56.1
Host is up (0.00040s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE        VERSION
135/tcp   open  msrpc          Microsoft Windows RPC
139/tcp   open  netbios-ssn    Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds?
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovsk:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
| smb2-security-mode:
|_  3.1.1:
|_    Message signing enabled and required
|_  smb2-time:
|_    date: 2026-09-21T05:48:28
|_    start_date: N/A
|_  clock-skew: 2s

TRACEROUTE (using port 80/tcp)
HOP RTT      ADDRESS
1   0.26 ms  192.168.56.1

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 28.19 seconds
```

Posiblemente con el escaneo que hicimos con -A obtuvimos mas informacion de los tipos de servicios y hasta de cierta manera un pequeño scrip con un -O igual hay otros que requieren muchisimo mas tiempo, pero siento que aun me falta ver la manera en como me funcionaria mas una busqueda. La Verdad es que si es muy interensante analizar que informacion te arroja cada comando y para que lo vamos a utilizar.